

Chapter 3

Polynomials and Their Shape

In the first two weeks we studied functions by looking for shape: lines have a fixed rate of change, parabolas have a vertex and symmetry, and transformations move known parent graphs into new positions. This week extends that same way of thinking to polynomials of higher degree.

A polynomial of high degree can look intimidating in expanded form, just a long sum of powers with no obvious shape. The main point of the week is that the graph is not as mysterious as it first appears. A small amount of information controls the rough shape: what the graph does at the far left and far right, and what happens at each root. The factored form displays this information. Each factor names a root, the exponent on that factor governs whether the graph crosses the axis or turns back from it, and the leading term governs both ends.

3.1 What a Polynomial Is

Definition: Polynomial. A *polynomial* is a function of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where the coefficients a_i are real numbers and $a_n \neq 0$. The *degree* is n , the highest power present; a_n is the *leading coefficient* and a_0 is the constant term.

Lines are degree-one polynomials and parabolas are degree-two, so this chapter is not beginning a completely new subject. It is widening a pattern we have already seen. A polynomial is built from powers of x , and the highest power present is its degree.

The expanded form above is convenient for evaluation and for reading the constant term $a_0 = f(0)$. If someone asks for $f(2)$, expanded form is often efficient. But expanded form can conceal the shape of the graph. The factored form reveals the places where the graph meets the x -axis, and those places are the anchors for a sketch.

Key idea. The expanded form of a polynomial is built for arithmetic; the factored form is built for shape. Where the graph crosses the axis, where it turns back, and where it goes at the two ends are all read from the factored form together with the leading term.

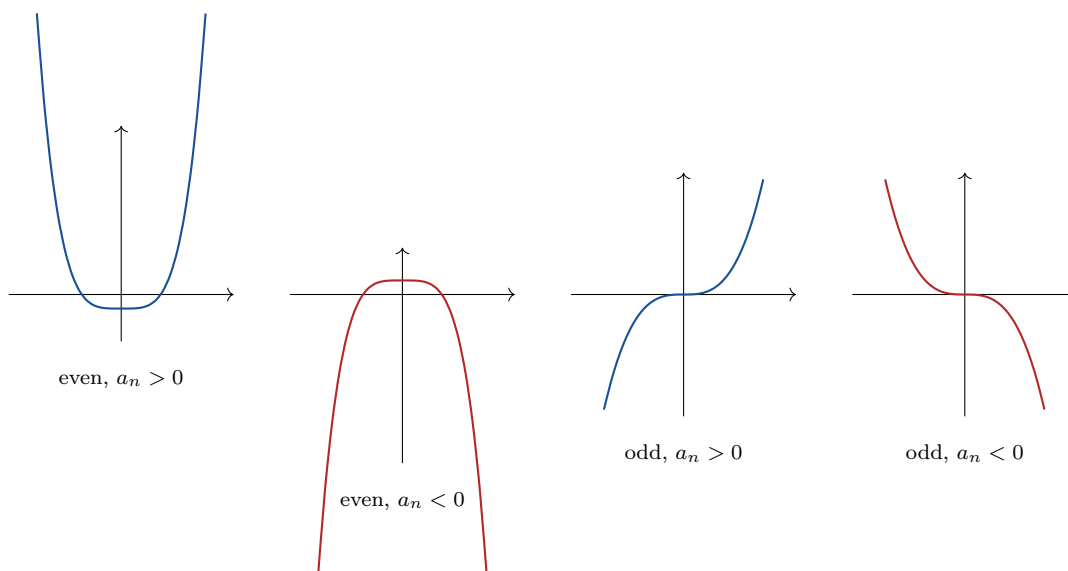
3.2 End Behavior: the Two Ends

When sketching a polynomial, it is natural to begin with the two ends of the graph. The ends tell us the large-scale direction of travel before we worry about the middle.

Far from the origin, the highest-degree term dominates every other term, because a high power outgrows all lower powers once $|x|$ is large. So the two ends of a polynomial's graph behave like the graph of its leading term $a_n x^n$ alone. Two features of that term decide the ends: the parity of the degree n and the sign of the leading coefficient a_n .

Key idea. End behavior of $f(x) = a_n x^n + \dots$

- *Even degree.* Both ends point the same way: up if $a_n > 0$, down if $a_n < 0$. The graph resembles a parabola at large scale.
- *Odd degree.* The ends point opposite ways: down-left to up-right if $a_n > 0$, up-left to down-right if $a_n < 0$. The graph resembles a cubic at large scale.



Example 3.2.1. (*reading the ends*) The polynomial $f(x) = -2x^5 + 7x^2 - 4$ has odd degree 5 and negative leading coefficient. The ends therefore run up on the left and down on the right. No factoring is needed; the leading term $-2x^5$ settles both ends by itself.

This does not mean the lower-degree terms are irrelevant. They can create turns, roots, and vertical shifts in the middle of the graph. They simply cannot compete with the leading term far out on the left and right.

Common pitfall. End behavior depends only on the leading term, never on the constant term or the middle coefficients. A large constant shifts the graph vertically but does not change where the ends go. Reading a_0 for end behavior is a common error; read $a_n x^n$.

3.3 Roots and the Factored Form

End behavior tells us where the graph is going at the two extremes. Roots tell us where the graph is pinned to the x -axis. These are the points where the output is zero, so they are also the places where the sign of the function may change.

A *root* (or zero) of f is an input r with $f(r) = 0$, a point where the graph meets the x -axis. The factored form lists the roots directly:

$$f(x) = a_n(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots (x - r_k)^{m_k}.$$

Each r_i is a root, and the exponent m_i is its *multiplicity*, the number of times that factor is repeated. The multiplicities sum to the degree n . This count is useful because it connects the local behavior at the roots to the total degree of the polynomial.

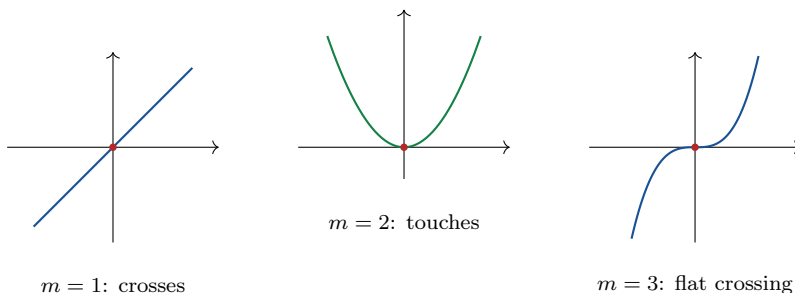
Definition: Multiplicity. The *multiplicity* of a root r is the exponent on the factor $(x - r)$ in the factored form. A root with multiplicity one is *simple*; higher multiplicities are *repeated*.

Multiplicity controls how the graph meets the axis at the root. The rule is a parity rule, the same parity idea that governed the ends. Odd powers preserve the sign of a factor as it passes through zero, while even powers remove the sign change. That algebraic fact is what becomes crossing or touching in the graph.

Key idea. Behavior at a root of multiplicity m :

- m *odd*. The graph *crosses* the axis; the function changes sign there. ($m = 1$: a straight crossing. $m = 3, 5, \dots$: a crossing that flattens against the axis.)
- m *even*. The graph *touches* the axis and turns back; the function does *not* change sign there. ($m = 2$: a parabola-like bounce.)

The picture. At a root, the factor $(x - r)^m$ changes sign exactly when m is odd, because a negative number raised to an odd power stays negative while an even power makes it positive. Crossing the axis is a sign change; touching and turning back is not. The multiplicity's parity is the whole story.



Example 3.3.1. (*reading roots and multiplicities*) For $f(x) = (x+2)(x-1)^2(x-3)$, the roots are -2 , 1 , and 3 . The roots -2 and 3 are simple (odd multiplicity), so the graph crosses there. The root 1 has multiplicity two (even), so the graph touches the axis at $x = 1$ and turns back without crossing. The degree is $1 + 2 + 1 = 4$, even, with positive leading coefficient, so both ends point up.

This one example already contains the basic sketching data: two ends, three roots, and the crossing-or-touching behavior at each root. A more accurate graph would need more information, but a faithful rough sketch begins here.

Interactive. Build a polynomial by placing roots and setting each multiplicity, in the form $y = (x - r_1)^{m_1} \cdots (x - r_n)^{m_n}$, then fit a hidden target. The fit score weighs root locations, multiplicities, and overall agreement. Watch how an even multiplicity makes the curve bounce off the axis while an odd multiplicity sends it through.

https://bobovski66.github.io/Precalculus/demos/polynomial_roots.html

3.4 The Sign Chart

The next step is to decide where the graph lies above the x -axis and where it lies below. Roots mark possible changes, but they do not by themselves tell us the sign on each interval. A sign chart supplies that information.

The roots are the only places a polynomial can change sign, because a polynomial is continuous and changes sign only by passing through zero. Between consecutive roots the function holds a single sign throughout. This is the observation that turns local data into a global sketch: mark the roots on a number line, and the line breaks into intervals on each of which f is entirely positive or entirely negative. One test point per interval reveals which.

Key idea. Sign chart procedure.

1. Factor f and mark all roots on a number line; they divide it into intervals.
2. Pick one test point in each interval and evaluate the sign of f there (the sign of the product of factors).
3. Label each interval $+$ or $-$. The sign can change only at a root, and only at a root of odd multiplicity.

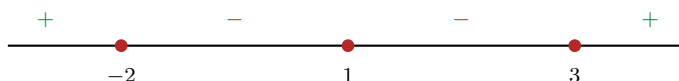
Example 3.4.1. (*building a sign chart*) Analyze $f(x) = (x + 2)(x - 1)^2(x - 3)$.

We already know the roots and multiplicities. Now we use the same factored form to track the sign of the product. The roots $-2, 1, 3$ divide the line into four intervals. Evaluate the sign of each factor at a test point in each interval; the factor $(x - 1)^2$ is never negative, so it never affects the sign.

Interval	test x	$(x + 2)$	$(x - 1)^2$	$(x - 3)$	sign of f
$(-\infty, -2)$	-3	$-$	$+$	$-$	$+$
$(-2, 1)$	0	$+$	$+$	$-$	$-$
$(1, 3)$	2	$+$	$+$	$-$	$-$
$(3, \infty)$	4	$+$	$+$	$+$	$+$

The sign stays negative across $x = 1$, confirming the even multiplicity: no sign change there. It flips at $x = -2$ and at $x = 3$, the simple roots.

The table may look like bookkeeping, but it is doing geometric work. Each plus sign places the graph above the axis on an entire interval, and each minus sign places it below.



The picture. The sign chart is a number line with the roots marked as special points and a sign assigned to each interval between them. It records, at a glance, where the function lives above the axis and where below. The graph must obey it: positive intervals hold the curve above the axis, negative intervals below, and the roots pin it to the axis.

3.5 Assembling the Global Sketch

We can now assemble the sketch. The point is not to plot dozens of values. The point is to combine the structural information already available from the formula.

End behavior fixes the two ends. The sign chart fixes which side of the axis the curve occupies on every interval. Multiplicity fixes how it meets each root. These three pieces of data determine the global shape up to the heights of the turns, which precalculus does not pin down exactly but calculus does. A faithful sketch needs only these pieces.

Key idea. Sketching a polynomial from its factored form.

1. Read the degree and leading coefficient; draw the two ends.
2. Mark the roots; at each, cross (odd multiplicity) or touch (even).
3. Use the sign chart to keep the curve on the correct side of the axis between roots.
4. Connect smoothly, consistent with the ends and the sign chart.

Example 3.5.1. (*a full sketch*) Sketch $f(x) = (x + 2)(x - 1)^2(x - 3)$.

We gather the information in the order that makes the graph easiest to draw: ends, roots, sign chart, then a smooth connection.

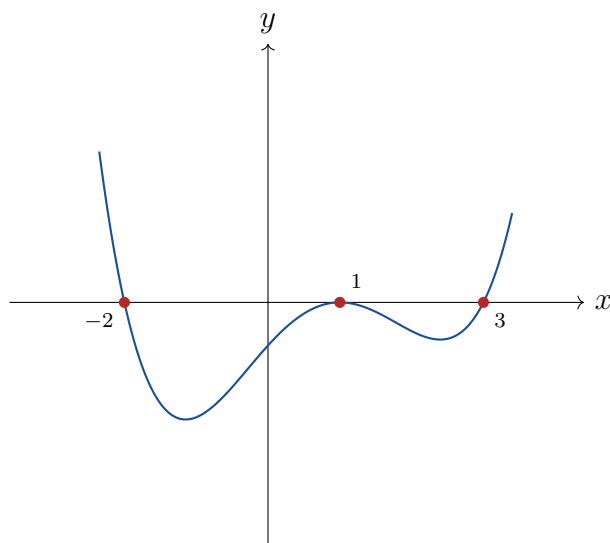
Ends. Degree 4, positive leading coefficient: both ends up.

Roots. Cross at -2 , touch at 1 , cross at 3 .

Sign chart. Positive on $(-\infty, -2)$, negative on $(-2, 3)$ except pinned to zero at $x = 1$, positive on $(3, \infty)$.

The curve descends from the upper left, crosses down at -2 , runs below the axis, rises to touch zero at $x = 1$ and falls back below, then crosses up at 3 and climbs to the upper right. The touch at $x = 1$ is the visual signature of the double root.

A sketch like this is not claiming exact turning-point heights. It is claiming the correct global structure: the right ends, the right intercept behavior, and the right side of the axis on every interval.



Common pitfall. The sign chart constrains which side of the axis the curve is on, not how high it rises. Without calculus the exact heights of the peaks and valleys are not determined, so sketches should show the correct *shape* (crossings, touches, sides, ends) and not claim exact turning-point heights.

3.6 Exercises

Exercises 3.6

- 3.6.1. State the end behavior of each polynomial from its leading term alone:
 (a) $f(x) = 3x^4 - x + 5$; (b) $f(x) = -x^3 + 2x^2$; (c) $f(x) = -4x^6 + x^5$; (d) $f(x) = x^5 - 3x^2 + 1$.
- 3.6.2. Explain why the constant term and middle coefficients have no effect on end behavior, referring to how powers compare when $|x|$ is large.
- 3.6.3. For $f(x) = (x + 1)(x - 2)^3(x - 4)$, list the roots with their multiplicities, state the degree, and say at each root whether the graph crosses or touches.
- 3.6.4. Using the parity of the exponent, explain why a factor $(x - r)^m$ changes sign at r when m is odd and does not when m is even.
- 3.6.5. Build a complete sign chart for $f(x) = (x + 3)(x - 1)(x - 2)$, with a test point in each interval and the sign of every factor recorded.
- 3.6.6. Build a sign chart for $f(x) = -(x + 1)^2(x - 3)$. Identify where the sign does and does not change, and connect each to a multiplicity.
- 3.6.7. Sketch $f(x) = (x - 1)(x + 2)^2$ from its factored form: state the ends, mark the roots with crossing or touching behavior, use the sign chart, and draw a shape consistent with all three.
- 3.6.8. Sketch a degree-five polynomial with a simple root at -2 , a triple root at 0 , a simple root at 3 , and positive leading coefficient. Describe the behavior at the triple root in words.
- 3.6.9. A polynomial has even degree and negative leading coefficient, with simple roots at -1 and 4 and a double root at 2 . Determine the lowest possible degree, then sketch a graph consistent with all stated features.
- 3.6.10. A polynomial is positive on $(-\infty, -1)$, negative on $(-1, 2)$, and positive on $(2, \infty)$, with no other sign changes. Determine the multiplicity parity at each

of -1 and 2 , and give a possible factored form of lowest degree.

3.6.11. Two students draw graphs of $f(x) = (x - 1)^2(x - 4)$ that differ only in the height of the turning point between the roots. Explain why both can be correct sketches and what additional tool would be needed to fix the height.

3.6.12. Open the Roots and Multiplicity Lab. Build a polynomial of degree at least four with at least one repeated root, record its factored form and the fit score against a target, and write one sentence on how changing a multiplicity from even to odd altered the graph at that root.